INCH-POUND
MIL-PRF-3098/54F
9 December 2004
SUPERSEDING
MIL-PRF-3098/54E
27 August 1997

PERFORMANCE SPECIFICATION SHEET

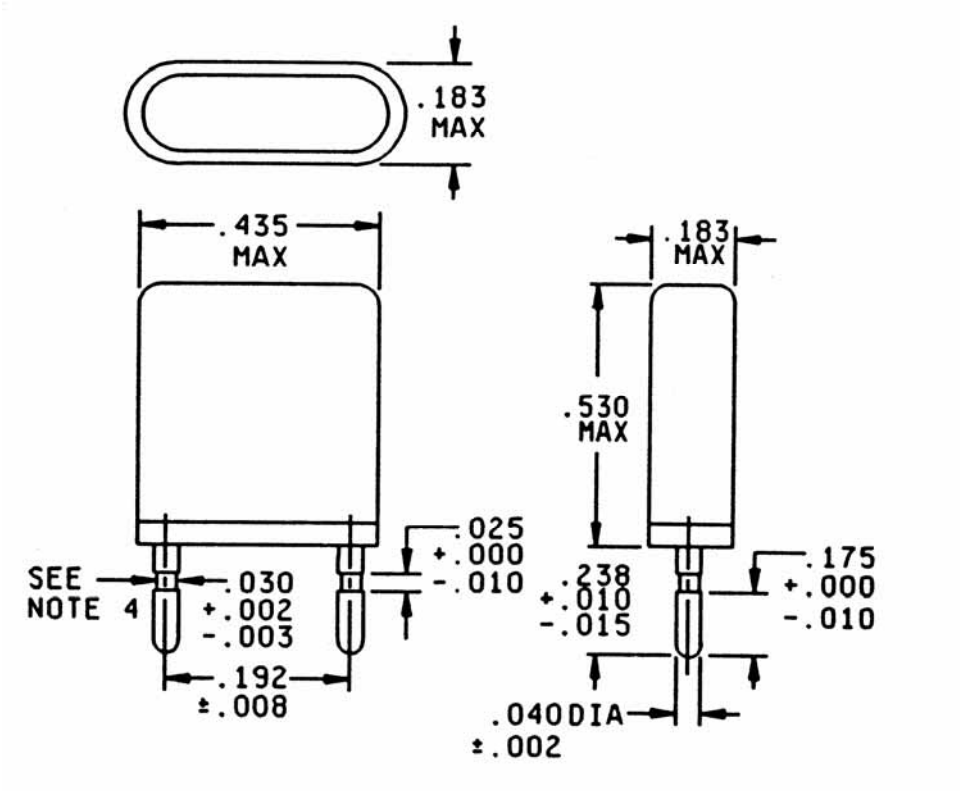
CRYSTAL UNIT, QUARTZ, CR73/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-3098.

Pertinent characteristics: 17 MHz to 61 MHz; third mechanical overtone; noncontrolled; series resonance.

Inches	mm
.002	0.05
.003	0.08
.008	0.20
.010	0.25
.015	0.38
.025	0.64
.030	0.76
.040	1.02
.175	4.45
.183	4.65
.192	4.88
.238	6.05
.435	11.05
.530	13.34



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-PRF-3098.
4. The pin undercut may be omitted.

FIGURE 1. Crystal unit - CR73/U.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 17.0 MHz to 61.0 MHz, inclusive.

Frequency tolerances: ±30 parts per million (ppm).

Rated drive level: 1.0 mW, maximum.

Resonance: Series.

Equivalent resistance: 17 MHz to 30.0 MHz - 40 ohms, maximum and 10 ohms, minimum.
30.0+ MHz to 61 MHz - 30 ohms, maximum and 5 ohms, minimum.

Unwanted modes: Method II, twice main mode resistance.

Mode of oscillation: Third mechanical overtone.

Operating temperature ranges: -55°C to +105°C, inclusive.

Capacitance, shunt: 17 MHz to 30 MHz - 3.5 pF ±0.5 pF.
30.0+ MHz to 61.0 MHz - 4.0 pF ±0.5 pF.

Shock (specified pulse):

Frequency change permitted: ±3 ppm.

Equivalent resistance change permitted: ±15 percent.

Thermal shock:

Frequency change permitted: ±3 ppm.

Equivalent resistance change permitted: ±15 percent.

Vibration: Method 204 of MIL-STD-202, test condition A.

Frequency change permitted: ±3 ppm.

Equivalent resistance change permitted: ±15 percent.

Aging:

Frequency change permitted: ±3 ppm.

* Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202

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* The margins of this specification sheet are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AR, MI
Navy - AS, MC, SH
Air Force - 19

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(Project 5955-0783)

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